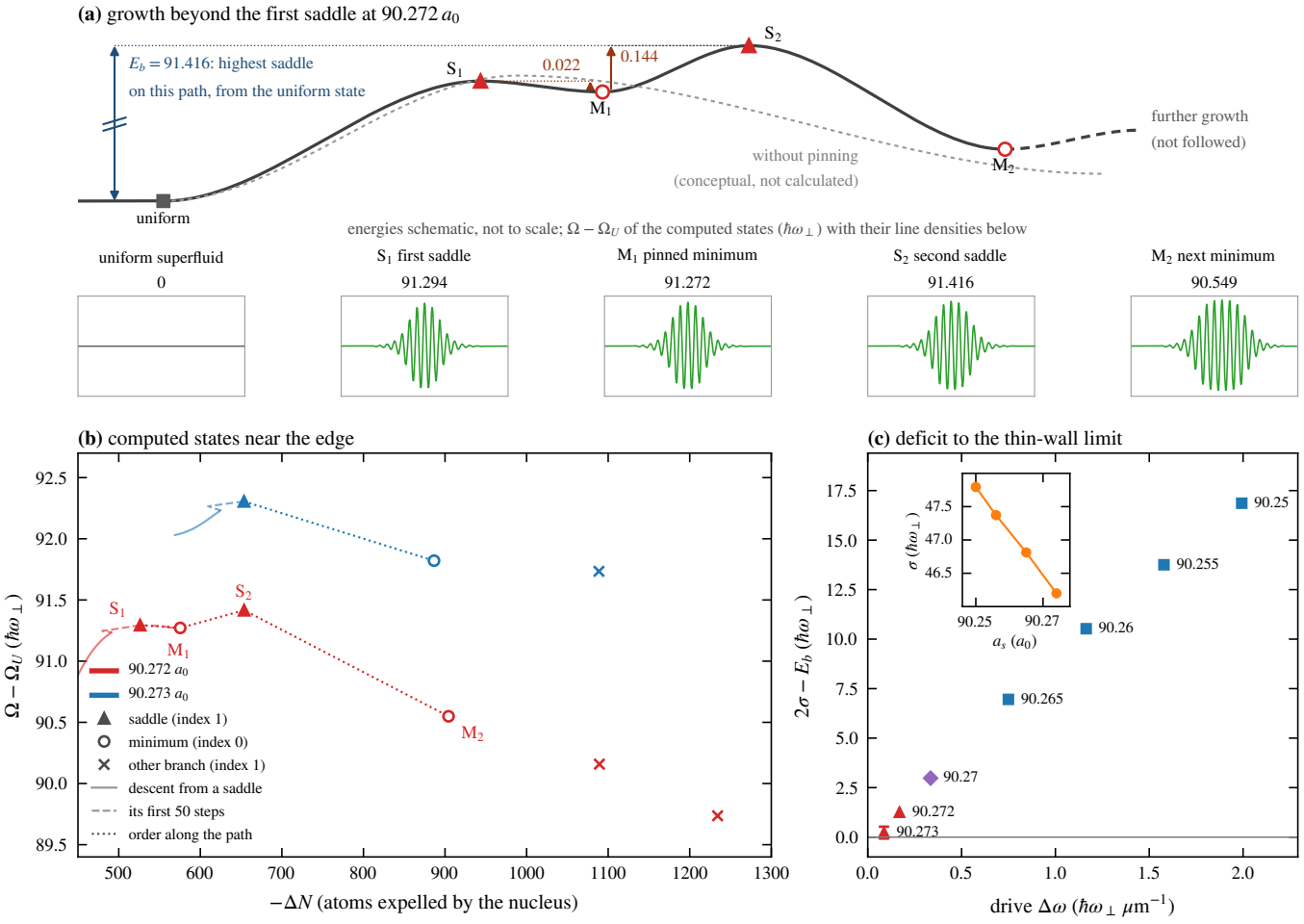
(a) growth beyond the first saddle at 90.272 $a_0$
$E_b = 91.416$: highest saddle
on this path, from the uniform state
S1
S2
M1
M2
0.022
0.144
uniform
without pinning
(conceptual, not calculated)
further growth
(not followed)
energies schematic, not to scale; $\Omega - \Omega_U$ of the computed states ($\hbar\omega_\perp$) with their line densities below
uniform superfluid
0
S1 first saddle
91.294
M1 pinned minimum
91.272
S2 second saddle
91.416
M2 next minimum
90.549
(b) computed states near the edge
$\Omega - \Omega_U$ ($\hbar\omega_\perp$)
$-\Delta N$ (atoms expelled by the nucleus)
90.272 $a_0$
90.273 $a_0$
saddle (index 1)
minimum (index 0)
other branch (index 1)
descent from a saddle
its first 50 steps
order along the path
(c) deficit to the thin-wall limit
$2\sigma - E_b$ ($\hbar\omega_\perp$)
drive $\Delta\omega$ ($\hbar\omega_\perp\,\mu\mathrm{m}^{-1}$)
$\sigma$ ($\hbar\omega_\perp$)
$a_s$ ($a_0$)
90.25
90.255
90.26
90.265
90.27
90.272
90.273